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Multimodal Deep Learning for Comprehensive Rock Lithology Characterization: A Novel Approach Combining Images and Formation Data

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Abstract

The presented technology is part of a broader AI-driven program focused on automating geological and petrophysical interpretation from primary rock samples images. Our solution extends this paradigm to drill cuttings—an underutilized yet abundant data source. We introduce a modular system of three ML-powered components—Image QC, Annotation QC, and Multimodal Lithology Classification. This end-to-end pipeline transforms raw, heterogeneous cuttings imagery into standardized, high-confidence training-ready datasets.

By combining image-based features with well data, the system enables consistent, scalable, and cost-effective prediction of lithology. Trained models are fully integrated into a user-friendly application, enabling geologists to interact with the system intuitively and perform real-time inference without coding. Ultimately, the approach increases data utilization, reduces expert time, and improves the accuracy and reproducibility of subsurface interpretation.

Introduction

The presented technology is part of a broader AI-driven program focused on automating geological and petrophysical interpretation from primary rock samples images. Our solution extends this paradigm to drill cuttings—an underutilized yet abundant data source. We introduce a modular system of three ML-powered components—Image QC, Annotation QC, and Multimodal Lithology Classification. This end-to-end pipeline transforms raw, heterogeneous cuttings imagery into standardized, high-confidence training-ready datasets.

By combining image-based features with well data, the system enables consistent, scalable, and cost-effective prediction of lithology. Trained models are fully integrated into a user-friendly application, enabling geologists to interact with the system intuitively and perform real-time inference without coding. Ultimately, the approach increases data utilization, reduces expert time, and improves the accuracy and reproducibility of subsurface interpretation.

Problem Statement for AI-based Cuttings Properties Prediction

Drill cuttings—loose rock fragments generated during drilling operations—represent one of the most accessible and abundant geological data sources at the wellsite. Despite their ubiquity, these materials are severely underutilized in geological interpretation workflows. Traditional lithological description from cuttings is labor-intensive, subjective, and highly dependent on expert availability. As a result, many projects rely on incomplete or inconsistent lithological logs, particularly when core sampling is sparse or unavailable.

In parallel with the success of AI models in core photo analysis, researchers and service providers have begun exploring machine learning (ML) applications for cuttings imagery. Studies by [Fu et al. \(2022\)](#), [Ding et al. \(2023\)](#), and [Marlot et al. \(2023\)](#) have demonstrated the feasibility of CNN-based lithology prediction from cuttings. However, these approaches typically assume curated, high-quality datasets and overlook the operational realities of image acquisition in field conditions.

Unlike core samples, cuttings are often photographed in trays or sample bags under inconsistent lighting, with variable resolution, non-geological artifacts (e.g., gloves, tools, labels), and noisy or mismatched lithology labels. These challenges severely impact the reliability and generalizability of ML models trained on such data. Existing literature and patents (e.g., US11443149B2, US12154324B1) offer limited solutions to these issues. Most focus on direct classification from RGB images while assuming ideal data conditions.

Proposed Solution and Novelty of the Present Work

This work introduces a comprehensive, modular pipeline for quality control and lithology classification tailored specifically to drill cuttings imagery. Unlike previous approaches, it directly addresses the visual variability, annotation inconsistency, and limited contextual information that characterize cuttings datasets.

The proposed system comprises three interlinked modules:

- ResNet-Based Image Quality Control (Image QC): Filters out visually irrelevant or contaminated images using a CNN trained on a labeled dataset of acceptable vs. unacceptable cuttings photos. Common rejection criteria include overexposure, blur, presence of non-geological objects, or insufficient rock content.
- Embedding-Based Annotation Consistency (Annotation QC): Uses MobileNetV2-derived embeddings and Jensen-Shannon divergence to detect visually inconsistent samples within lithology classes. This module flags potential mislabels, edge cases, or ambiguous examples for expert review, improving training data quality.
- Lithology Prediction (Multimodal Classification): Employs CNN-based classifiers (MobileNetV2 or ResNet variants) to predict lithology classes from image patches. A multimodal extension optionally integrates metadata—such as formation data—for improved accuracy in visually ambiguous cases.

All models are trained on expert-annotated field datasets and implemented in Python using Jupyter Notebooks for interactive development. The output models are further deployed within an inhouse user application, enabling geologists to upload raw images, and receive real-time lithology predictions without coding.

This approach offers several innovations over prior work:

- Dedicated, reproducible QC stages embedded into the ML workflow
- Semi-automated label auditing via embedding divergence
- Support for both image-only and metadata-augmented classification
- Scalable architecture, adaptable to new fields and formations

Together, these components form a robust, field-ready system for unlocking the geological value hidden in cuttings datasets. By improving data quality, model performance, and workflow consistency, the solution enables broader use of cuttings imagery in AI-assisted reservoir characterization.

Pipeline Background: ML Algorithms and Their Applications

The lithology analysis pipeline for drill cuttings imagery integrates three key machine learning components: ResNet-50-based visual quality control, embedding-based annotation consistency analysis, and a MobileNetV2-based lithology classification model. Each module is tailored to a specific stage of the workflow, leveraging the strengths of deep convolutional architectures for visual inspection, label auditing, and automated lithological prediction. Together, they enable a scalable and robust system for high-throughput cuttings image interpretation.

ResNet-50: Visual Quality Control for Cuttings Image Classification

Principle. Residual Networks (ResNets), introduced by He et al. (2015), are deep convolutional neural network (CNN) architectures designed to mitigate the vanishing gradient problem in very deep networks. ResNet-50, a 50-layer variant, utilizes identity-based skip connections to learn residual mappings instead of direct mappings. By allowing gradients to flow more directly through shortcut connections, ResNets enable the successful training of extremely deep architectures.

The core idea is to reformulate the desired mapping $H(x)$ as a residual function $F(x)=H(x)-x$, so the network learns $F(x)+x$. This structure allows the network to model identity mappings easily when deeper layers are unnecessary, stabilizing training and improving accuracy.

Mathematical Formulation. Given an input feature map, a residual block compute:

$$\hat{y} = F(x, \{W_i\}) + x \quad (1)$$

Where $F(x, \{W_i\})$ represents the residual function (e.g., two or three convolutional layers with learnable weights) and x is the input that is added back to the output through an identity or projection shortcut. In ResNet-50, the architecture is composed of:

- An initial convolution and max-pooling layer,
- Followed by 16 residual blocks grouped into four stages,
- Each block consisting of a bottleneck design: 1×1 , 3×3 , and 1×1 convolution.

Illustrative diagram of the ResNet-50 architecture, emphasizing residual blocks with identity shortcuts, is presented in [Figure 1](#). The network consists of an initial convolutional block (Stage 1), followed by four stages of convolutional and identity residual blocks (Stages 2-5). Each residual block enables gradient flow through skip connections. The final layers include average pooling, flattening, and a fully connected layer for classification output.

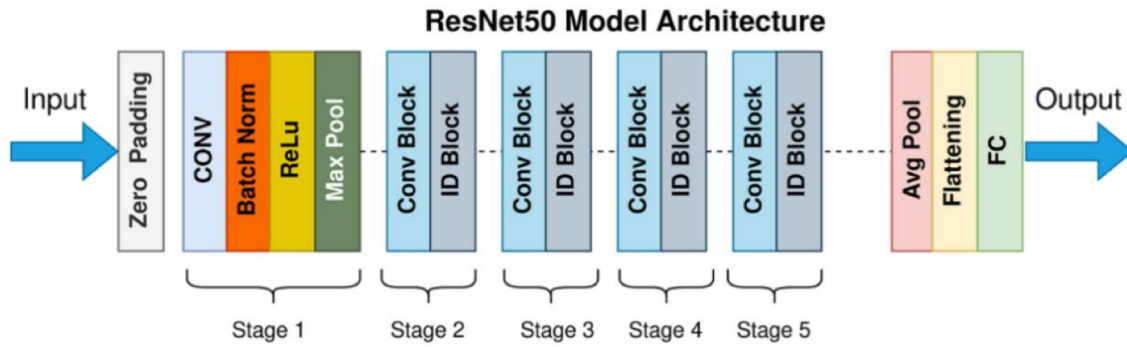


Figure 1—ResNet-50 model architecture.

Application Justification. In the context of filtering out irrelevant drill cuttings images, a multiclass classification model can be trained to distinguish between valid samples and various unwanted categories—such as images with excessive drilling mud, chemical additives, wet aggregates, or noisy lab backgrounds. Instead of relying solely on handcrafted statistical features, deep convolutional embeddings are extracted from each image using ResNet-50, pretrained on large-scale image datasets. These embeddings capture high-level semantic features such as texture, color composition, and structural patterns, which vary systematically across different image types. The ResNet-50 architecture, with its residual connections and deep representation capacity, enables effective learning of such distinctions, even in noisy, heterogeneous datasets typical of cuttings imagery.

Embedding-Based Analysis: Quality Control of Lithology Annotations

Principle. In this module, each cuttings image is represented by a precomputed feature embedding, obtained from an intermediate layer of a convolutional neural network such as ResNet-50/MobileNet/EfficientNet. These embeddings encode key aspects of visual appearance—including color distributions, texture patterns, and structural features—into a compact representation. Alternatively, RGB histograms or cumulative color distribution vectors can be used for lightweight approximation. To assess the consistency of expert-provided lithology labels, the embedding of each image is compared against the distribution of embeddings from other images assigned to the same lithology class. The underlying assumption is that visually coherent samples should form tight clusters in the embedding space.

Mathematical Formulation. Let $z_i \in \mathbb{R}^d$ denote the embedding of image i , and let $C_k = \{z_j | y_j = k\}$ be the set of embeddings for all images labeled as lithology class k . The class centroid μ_k is computed as:

$$\mu_k = \frac{1}{|C_k|} \sum_{z_j \in C_k} z_j \quad (2)$$

To evaluate the divergence of each embedding z_j from its class distribution, a symmetric divergence metric such as Jensen-Shannon divergence (JSD) is applied:

$$D_{JSD}(z_i | \mu_k) = \frac{1}{2} D_{KL}(z_i | m) + \frac{1}{2} D_{KL}(\mu_k | m) \quad (3)$$

Where $m = \frac{1}{2}(z_i + \mu_k)$ and D_{KL} is the Kullback-Leibler divergence.

Images with high JSD scores relative to their class centroid are flagged as potential annotation outliers or labeling errors, as they deviate significantly from the visual norm of their declared lithology. These may indicate mislabeling, edge cases, or inconsistent expert criteria.

Application Justification. Accurate lithology labeling is essential for training reliable ML models, but expert annotations of drill cuttings images are often inconsistent due to subjective judgment and varying image quality. The embedding-based QC approach provides an objective way to flag visually inconsistent

samples within each class by measuring divergence in feature space. This enables efficient detection of potential labeling errors, supports iterative label refinement, and improves the overall quality and reliability of the dataset for downstream geological interpretation.

MobileNetV2: Lightweight Lithology Classification from Cuttings Imagery

Principle. MobileNetV2, introduced by [Sandler et al. \(2018\)](#), is an efficient convolutional neural network architecture designed for mobile and resource-constrained environments. It incorporates two core innovations: inverted residual blocks and linear bottlenecks, which enable deeper networks with fewer parameters. By leveraging depth wise separable convolutions, MobileNetV2 significantly reduces computational cost while preserving competitive accuracy across vision tasks.

Mathematical Formulation:

An inverted residual block in MobileNetV2 consists of:

1. A 1×1 convolution with ReLU6 activation (expansion layer).
2. A 3×3 depth wise convolution with ReLU6 activation.
3. A 1×1 pointwise linear convolution (projection layer).

A residual (shortcut) connection is applied between the block's input and output when the input and output dimensions are equal. This structure allows the network to retain representational capacity while minimizing information loss. [Figure 2.1](#) and [Figure 2.2](#) illustrate the core architecture of a MobileNetV2 block.

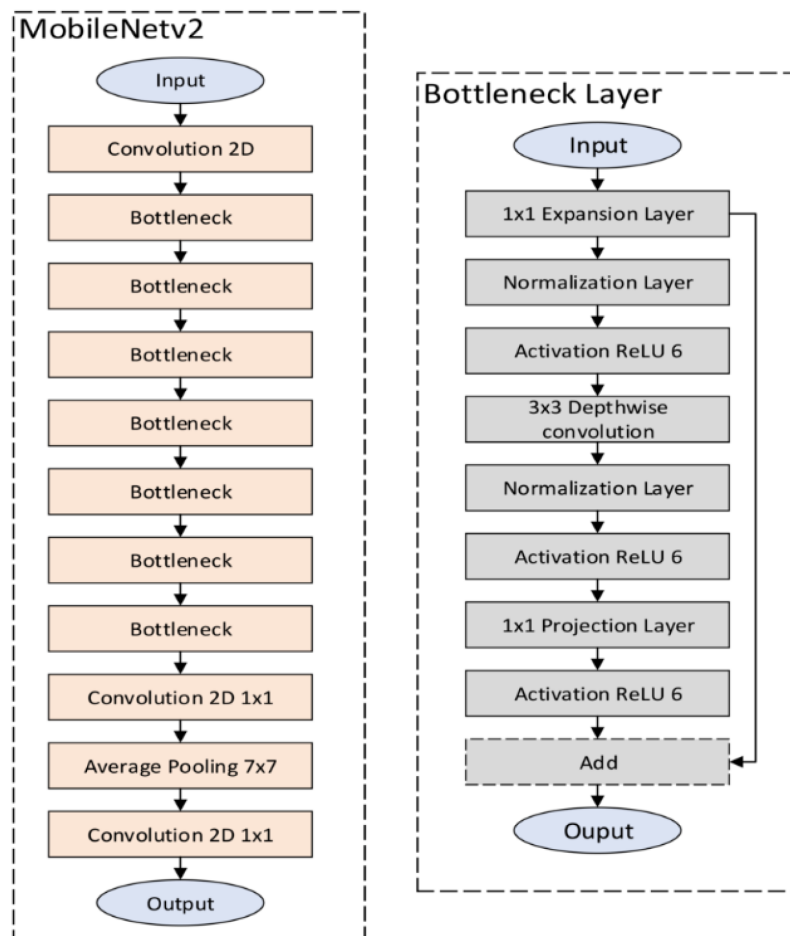


Figure 2.1—Architecture of MobileNetV2 neural network: [Tragoudaras \(top\) et al., 2022](#), [Akay et al., 2021](#) (bottom).

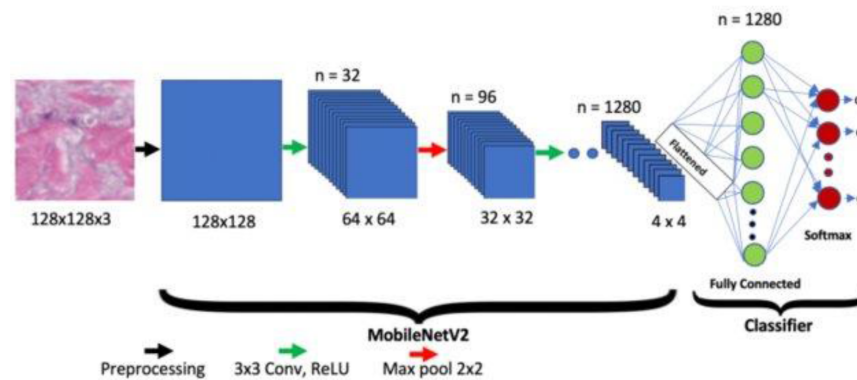


Figure 2.2—Architecture of MobileNetV2 neural network: [Tragoudaras \(top\) et al., 2022](#), [Akay et al., 2021](#) (bottom).

Application Justification. Drill cuttings images offer a cost-effective but underutilized source for lithology interpretation. MobileNetV2 provides an efficient solution for automated lithology classification based on visual patterns in cuttings photos. Its compact architecture enables fast inference on large datasets or edge devices, while still capturing key features such as grain texture, color distribution, and structural cues.

This makes it well-suited for scalable, real-time deployment in geological workflows where rapid, image-based lithology prediction is required.

Pipeline Customization for Cutting Description

The lithology prediction workflow for drill cuttings imagery is composed of three core machine learning (ML) modules, each addressing a distinct task in the end-to-end interpretation pipeline. In the initial phase, models are trained using expert-labeled image datasets, with each component optimized for a specific type of visual analysis. The integrated system supports automated screening, quality control, and lithology classification of cuttings photographs.

The key modules include:

- ResNet-50-based classification of visually irrelevant images, such as those containing excessive mud, background clutter, or chemical contamination;
- Embedding-based quality control of expert lithology labels through anomaly detection in feature space;
- MobileNetV2-based lithology prediction, enabling fast and scalable classification of cuttings images into predefined lithological categories.

Each module is fully configurable, with tunable hyperparameters and support for user-provided datasets.

ResNet-50-Based Visual Quality Control

This module performs automated filtering of visually irrelevant drill cuttings images that may reduce model performance if included in the training dataset. The module targets images contaminated by excessive drilling mud, chemical additives, labeling artifacts, or noisy lab backgrounds.

A ResNet-50 convolutional neural network was fine-tuned to perform multiclass classification across predefined noise categories and a valid class.

The training dataset was constructed from manually verified image sets, with class definitions informed by expert visual criteria. Input images were resized and normalized prior to training, and standard data augmentation techniques were applied to improve generalization. The final model achieved a mean F1-score of 0.85 across all classes.

The model is integrated into the preprocessing pipeline and automatically flags or excludes irrelevant images, significantly improving dataset consistency and robustness of the downstream lithology classification.

Step-by-Step Workflow.

Data annotation

The dataset is built from manually reviewed and pre-labeled images, categorized into classes such as "clean sample," "mud-contaminated," "background-noise," and others (see [Figure 2](#) for representative examples). The pipeline is flexible and user can define extra classes if needed. These labels provide the supervision necessary to train a robust multi-class image classifier.



Figure 2—Raw cuttings images classified as "mud-contaminated" (left) and "clean" (right).

Data Loading and Structure Setup

Images are loaded from a predefined directory. During preprocessing, output subfolders are automatically created for storing:

- Orientation-corrected images,
- Size-validated inputs,
- Training logs and saved models,
- Labeled test cases categorized as good or faulty.

Initial binary or multiclass labels are attached to each image based on expert review metadata.

Feature Generation

Image embeddings are extracted using a pretrained ResNet-50 model. The input images are resized and normalized, and high-level feature vectors are obtained from intermediate network layers. These embeddings capture spatial structure, color consistency, and texture—critical for identifying irrelevant or noisy imagery. Faulty or unreadable inputs are filtered during this step.

Model Training

The extracted embeddings are used to train a supervised classifier. In this implementation, ResNet-50 is fine-tuned for multi-class classification. Alternatively, embeddings can be passed to a shallow classifier (e.g., Random Forest) for faster experimentation. Key metrics such as accuracy, precision, recall, and confusion matrices are computed.

Evaluation

The model consistently achieves high classification performance, with average precision and recall above 0.85 across all irrelevant image categories. The confusion matrix analysis confirms low false-positive rates for clean sample detection, ensuring minimal over-filtering.

Saving Results

Once trained, the model is serialized and stored in the project output directory. It is then could be integrated into the preprocessing pipeline within the inhouse platform, where it performs real-time screening of uploaded images. Irrelevant samples are automatically flagged or excluded, improving the consistency and reliability of the lithology training dataset (Figure 3).

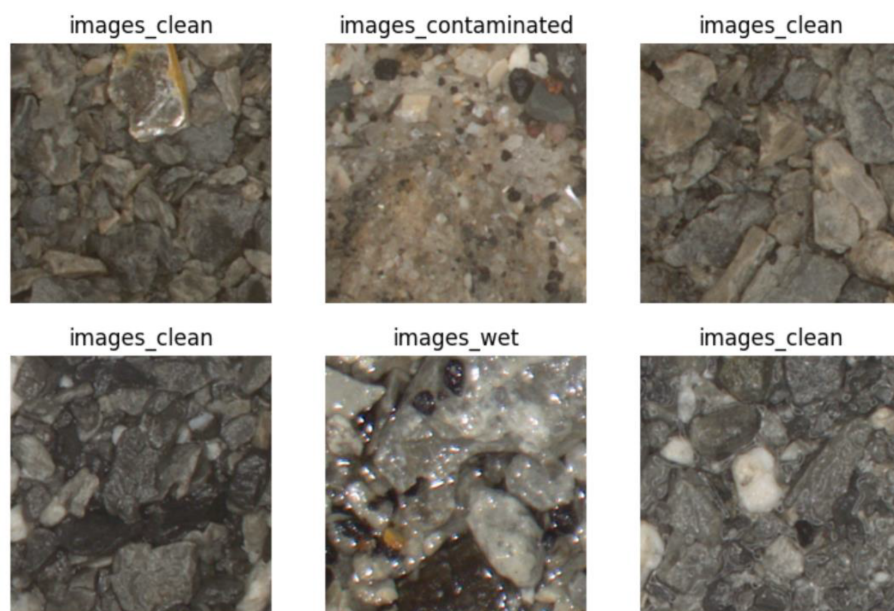


Figure 3—Results of trained QC model prediction.

Embedding-Based Quality Control of Lithology Annotations

This module provides an automated method for auditing expert-provided lithology labels based on visual similarity in embedding space. For each image, feature embeddings are extracted from a mid-layer of a pretrained CNN model. These embeddings capture texture, color distribution, and structural appearance. The system computes the Jensen-Shannon divergence between each image embedding and the centroid of its declared lithology class. Samples with high divergence are flagged as visual outliers, which may indicate annotation inconsistencies, borderline cases, or mislabeling. The flagged cases are surfaced for expert review or iterative relabeling.

This unsupervised validation tool enhances labeling reliability, supports curation of clean datasets, and increases confidence in training data used for supervised lithology prediction.

Step-by-Step Workflow. This module provides automated validation of expert-assigned lithology labels by analyzing visual consistency within each lithology class in the embedding space. It helps identify potentially mislabeled or ambiguous images that may compromise supervised model training if left unchecked.

Data Loading and Label Grouping

Expert-labeled cuttings images are loaded from the project dataset. Each image is associated with a lithology class (e.g., sandstone, shale, dolomite, etc.). The dataset is grouped by declared class, allowing within-class comparison of visual representations.

Embedding Extraction

For each image, feature embeddings are extracted using a pretrained CNN model or RGB vectors (Figure 4). Embeddings are typically taken from a high-level intermediate layer and represent each image as a fixed-length vector in feature space. These embeddings encode complex visual characteristics such as texture, grain size, and structural composition.

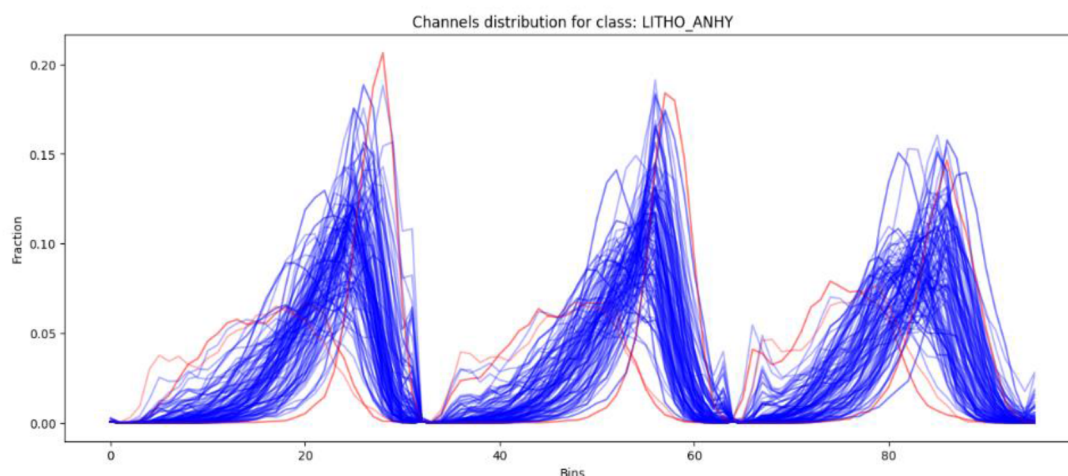


Figure 4—Result of Expert label validation: red - outliers, blue - consistent samples.

Divergence Computation

Within each lithology class, the centroid of the embedding cluster is computed as the mean vector across all valid samples. For every image in the class, Jensen-Shannon divergence (JSD) is calculated between its embedding and the class centroid. This divergence quantifies visual dissimilarity relative to the class average.

Outlier Detection

Images with divergence above a configurable threshold (e.g., 95th percentile) are flagged as potential annotation outliers. These may represent:

- Incorrectly labeled samples,
- Visual edge cases,
- Interclass ambiguity,
- Inconsistent expert interpretation.

The flagged samples are automatically visualized and stored in a separate output directory for expert review and relabeling. Anomaly statistics are logged for each class.

Review and Integration

The output includes a ranked list of suspicious annotations, annotated plots of embedding distributions, and per-class divergence histograms. This information supports targeted dataset curation and reduces noise in the training labels.

The module is designed to operate with minimal configuration and can be reused across projects with different lithology schemes or image resolutions.

MobileNetV2-Based Lithology Prediction

This module enables automated classification of cuttings images into lithology classes using a lightweight convolutional neural network. A MobileNetV2 backbone was selected for its efficiency and suitability for deployment in constrained environments or large-scale batch processing.

The model was trained on a curated dataset with expert-assigned lithology labels. Each image was randomly cropped to several 512×512-pixel patches and passed through a MobileNetV2 network with a global average pooling layer and a fully connected softmax output. The model achieved a validation accuracy of 87% across seven lithology classes (e.g., sandstone, shale, dolomite, limestone, siltstone, argillite, anhydrite) and demonstrated robustness to visual variability in field conditions.

Once trained, the model was exported and embedded in the application, enabling real-time lithology inference from uploaded cuttings photos.

Step-by-Step Workflow. This module implements supervised lithology classification based on drill cuttings imagery using a MobileNetV2-based convolutional neural network. Its lightweight architecture enables scalable training and deployment, including on edge devices or within high-throughput pipelines.

Data Preparation

Raw cuttings images are cropped into multiple 512×512-pixel patches using a randomized sampling strategy with quality control filters. This ensures exclusion of overexposed, underexposed, or structurally invalid samples. Each patch inherits the lithology label from its corresponding full image, based on expert-verified metadata.

Dataset and Augmentation

The model is trained on a balanced and preprocessed dataset using PyTorch Dataset and DataLoader abstractions. Data augmentation techniques such as random rotation, flipping, and normalization are applied to improve generalization. Additional crops are generated for underrepresented lithology classes to mitigate class imbalance.

Model Architecture

The classification model is built on a pretrained MobileNetV2 encoder. It includes a global average pooling layer followed by a fully connected classification head. The final softmax layer outputs a probability distribution across lithology classes.

Model Training

After assembling a curated dataset of cuttings images with verified lithology labels, the model training pipeline is launched to perform multi-class rock type classification.

- Input Dataset Preparation

Cropped image patches (512×512 pixels) are extracted from full-size cuttings images using a quality-controlled generator. The dataset is then split into training and validation sets using stratified sampling to preserve lithology class distribution. Minor classes are oversampled by generating additional crops to address class imbalance.

- Model Architecture

The classification network is based on a MobileNetV2 backbone, chosen for its computational efficiency and high representational power. The final architecture includes:

- A pretrained MobileNetV2 encoder (excluding classifier head),
- A global average pooling layer,
- A fully connected classification head with softmax activation.

- Training Configuration

- Batch size: 64
- Epochs: 15
- Loss function: focal
- Optimizer: Adam
- Scheduler: StepLR or ReduceLROnPlateau
- Metrics Tracked: Accuracy, Precision, Recall, F1-score (weighted)

- Augmentation Strategy

To improve generalization and mitigate overfitting, the training pipeline applies random horizontal and vertical flips, 90° rotations, and optional instance normalization. Figure 5 shows an example batch from the training dataset.

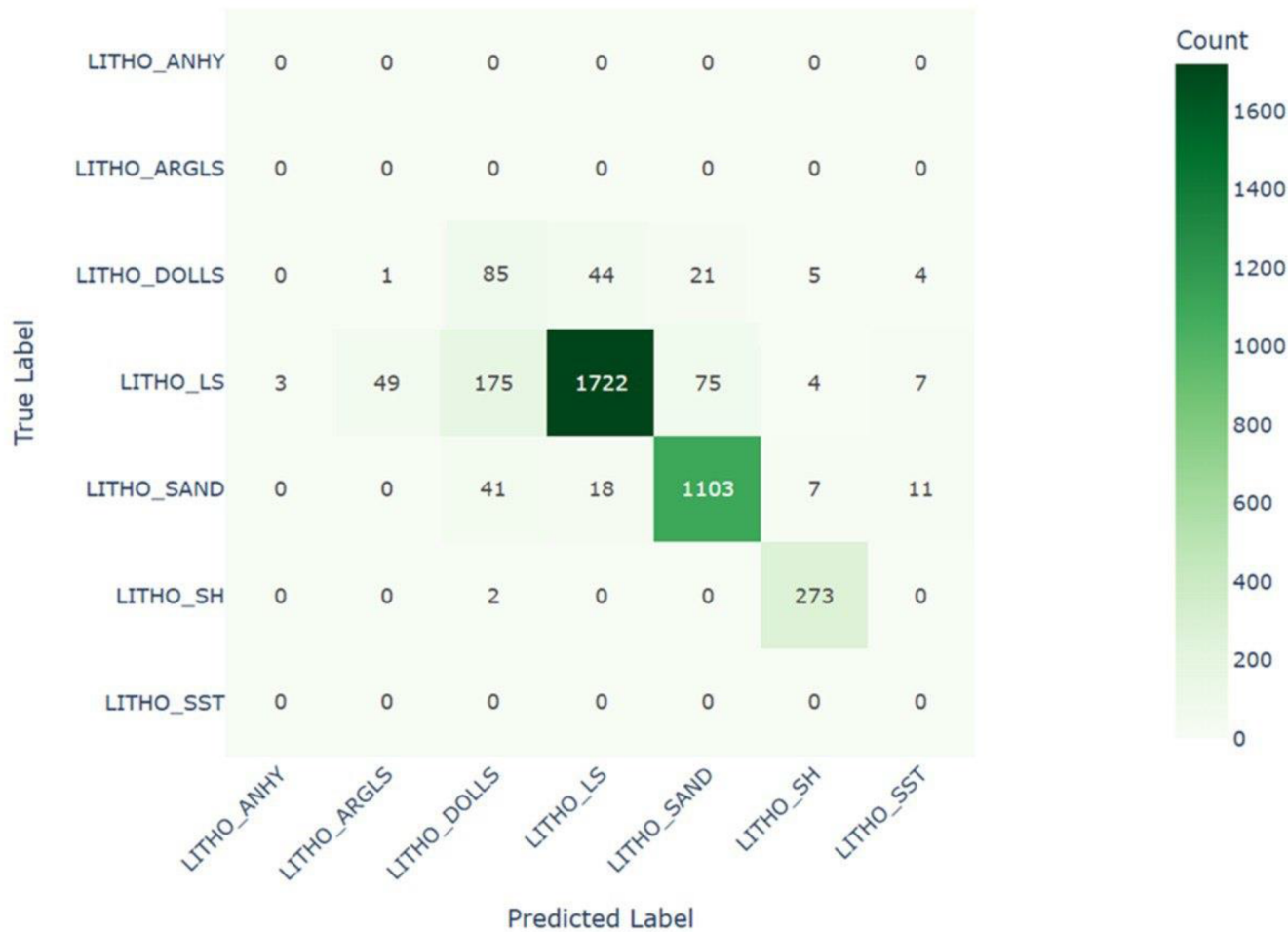


Figure 5—Confusion matrix for one of the testing runs.

Performance Evaluation

Model evaluation is conducted on a hold-out validation set. Key metrics are computed using the TorchMetrics library. On a 7-class lithology classification task (e.g., sandstone, shale, dolomite, limestone, siltstone, argillites, anhydrites), the model achieved:

- Accuracy: 0.87
- Precision: 0.86
- Recall: 0.84
- F1-score: 0.85

Confusion matrix is saved for further analysis.

Saving Results and Deployment

The final model is exported in two formats:

- .pth — for integration with PyTorch inference pipelines;

- .onnx — for compatibility with production environments or edge devices.

The model is embedded within the inhouse platform, allowing real-time lithology prediction during interactive or automated cuttings image workflows.

User Application: Deployed Model for Lithology Prediction

The trained CNN model for lithology prediction has been successfully deployed within the inhouse cutting Cutting user interface, which has been specifically adapted to support machine learning workflows. The GUI enables users to upload drill cuttings images and perform real-time lithology inference using the integrated CNN-based model (Figure 6). This seamless integration ensures accessibility of advanced ML tools to domain experts without requiring programming expertise.

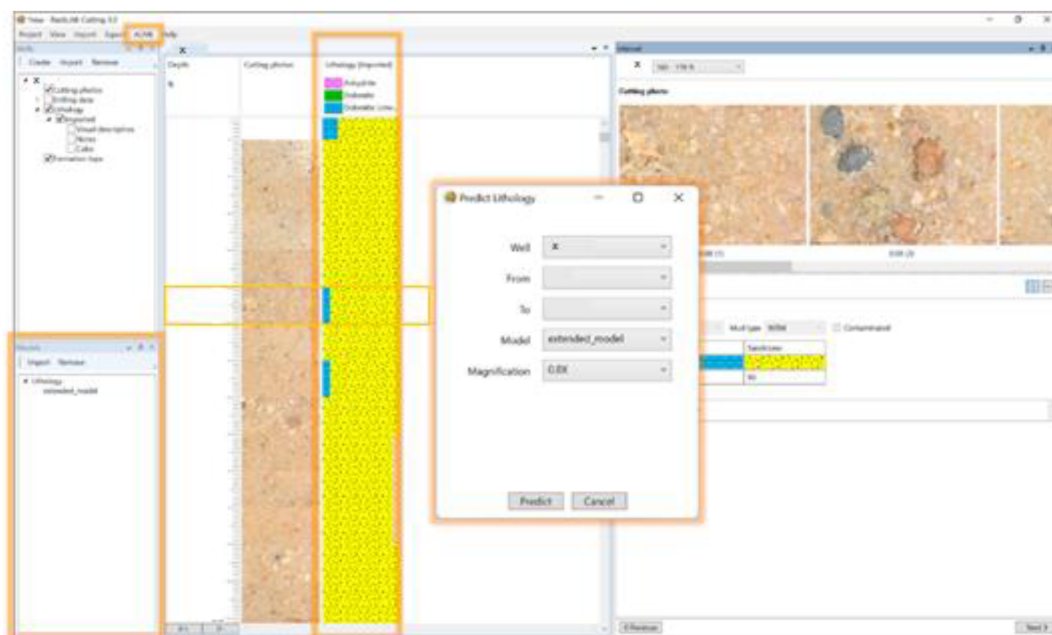


Figure 6—Inhouse Cutting user application with implemented ML model.

Conclusions

This paper introduces an innovative multimodal AI methodology for lithology prediction, combining image and encoded geological data. By integrating formation tops through target encoding, it advances the state of knowledge in petroleum engineering, providing a more comprehensive and reliable alternative to traditional and image-only approaches. The findings present a significant contribution to efficient reservoir characterization.

Initial experiments on a curated dataset demonstrated that incorporating formation tops embeddings significantly improved prediction metrics compared to image-only models. The multimodal model achieved higher classification accuracy and F1 scores, showcasing its ability to generalize effectively on unseen wells. By leveraging geological context, the model enhances the differentiation of visually similar lithotypes, providing a more comprehensive understanding of rock characteristics. This innovative approach reduces subjectivity and accelerates the traditionally time-consuming geological analysis process. The results highlight the method's robustness in predicting lithology, particularly in geologically complex scenarios, and underscore its potential for widespread adoption in petroleum engineering. This development marks a significant advancement in automated lithological classification, bridging the gap between data-driven methodologies and practical field applications. As the approach integrates geological context into predictive

modeling, it opens new avenues for improving efficiency and accuracy in subsurface exploration and reservoir characterization.

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